STRICT LOG-CONCAVITY OF INVERSE Z-POLYNOMIALS OF PAVING MATROIDS

KNOWLEDGE SENSE

Abstract. The inverse Z-polynomial $Y_M(t)$ of a loopless matroid M is a palindromic polynomial of degree $\operatorname{rk} M$ with nonnegative coefficients. A related polynomial was introduced earlier by Ferroni, Matherne, Stevens and Vecchi; we use the normalization of Gao, Ruan and Xie. Gao, Ruan and Xie conjectured that the coefficients form a log-concave sequence with no internal zeros, and proved this for sparse paving matroids. We prove that for every loopless paving matroid of rank $k \geq 2$ the coefficients of Y_M are strictly log-concave at every internal index. Separately, all coefficients of Y_M in degrees $0, \ldots, \operatorname{rk} M$ are positive for every loopless matroid M. The proof starts from the explicit formula of Gao, Li and Xie, which expresses Y_M through inverse Z-polynomials of uniform matroids and the numbers of stressed hyperplanes of each size. After normalization, the coefficient sequence is a convex combination of explicit sequences whose consecutive ratios lie in common, pairwise separated intervals; this settles odd rank. In even rank $k = 2m$ all components have the same increment at the central coefficient, which gives the middle inequality and settles $k = 2$. For $k = 2m$ with $m \geq 2$, the inequalities at the indices $m-1$ and $m+1$ require a separate lower bound on the coefficient of t^{m-1}; it uses the fact that a paving matroid on n elements has at most one hyperplane with more than $(n+k-2)/2$ elements.

1. Introduction

Let M be a loopless matroid. The Kazhdan–Lusztig polynomial $P_M(t)$ of Elias, Proudfoot and Wakefield [3], the inverse Kazhdan–Lusztig polynomial $Q_M(t)$ of Gao and Xie [8], and the Z-polynomial

$$Z_M(t) = \sum_{F \in \mathcal{L}(M)} t^{\operatorname{rk} F} P_{M/F}(t)$$

of Proudfoot, Xu and Young [10] have nonnegative coefficients [2], and their coefficient sequences are the subject of several conjectures on unimodality, log-concavity and real-rootedness. Within the Kazhdan–Lusztig–Stanley framework, Ferroni, Matherne, Stevens and Vecchi [4, p. 19, footnote 7] introduced a related polynomial built from Q_M, as an analogue of Z_M. Gao,

Date: 12 October 2026.

2020 *Mathematics Subject Classification.* Primary 05B35; Secondary 05A20.

Key words and phrases. Paving matroid, inverse Z-polynomial, inverse Kazhdan–Lusztig polynomial, log-concavity, stressed hyperplane.

Ruan and Xie [7] studied the *inverse Z-polynomial* $Y_M(t)$ in the normalization recalled in Definition 2.2, which we use throughout; it has nonnegative coefficients and satisfies $t^{\operatorname{rk} M} Y_M(t^{-1}) = Y_M(t)$. They computed Y_M for uniform and sparse paving matroids, proved that its coefficients are log-concave with no internal zeros when M is sparse paving [7, Theorem 1.6], and conjectured that this holds for every matroid [7, Conjectures 1.4 and 1.5]; see also [1, Conjecture 1.2(d)]. Gao, Li and Xie [6, Theorem 1.7] later gave an explicit formula for Y_M for all paving matroids, in a form different from the earlier formula [7, Proposition 5.3].

A matroid of rank k is *paving* if every circuit has at least k elements. Our main result settles the log-concavity question for this class, with strict inequalities.

Theorem 1.1. *Let M be a loopless paving matroid of rank $k \geq 1$ on a finite set of n elements, and write $Y_M(t) = \sum_{i=0}^{k} a_i t^i$. Then*

(i) *$a_i > 0$ and $a_i = a_{k-i}$ for $0 \leq i \leq k$;*
(ii) *if $k \geq 2$, then $a_i^2 > a_{i-1} a_{i+1}$ for $1 \leq i \leq k-1$.*

The symmetry in (i) holds for all matroids [7, Lemma 2.2], and positivity holds for all loopless matroids (Remark 2.3); our proof gives positivity for paving matroids independently. The new statement is (ii). It extends [7, Theorem 1.6] from sparse paving matroids to all paving matroids and shows that every internal inequality is strict. Matroids covered by Theorem 1.1 include uniform matroids, free matroids, and paving matroids with coloops or with several connected components; no representability assumption is made. Xie and Zhang [12] proved that the inverse Kazhdan–Lusztig polynomial Q_M of a paving matroid is log-concave. The polynomial Y_M is assembled from the polynomials $Q_{M|F}$ of the restrictions to all flats F, shifted and weighted by Möbius invariants of contractions (see (2.4)); the argument below does not use [12].

Corollary 1.2. *Under the hypotheses of Theorem 1.1, if $k \geq 2$, then $a_{i-1} < a_i$ for $1 \leq i \leq \lfloor k/2 \rfloor$ and $a_i = a_{k-i}$ for $0 \leq i \leq k$. In particular $a_{\lfloor k/2 \rfloor} = a_{\lceil k/2 \rceil}$, and $a_0, \ldots, a_k$ is unimodal.*

Proof. Put $r_i = a_i / a_{i-1}$ for $1 \leq i \leq k$. By Theorem 1.1 the sequence $r_1, \ldots, r_k$ is strictly decreasing, and $r_{k+1-i} = 1/r_i$ by symmetry. If $i \leq \lfloor k/2 \rfloor$, then $i < k+1-i$, hence $r_i > r_{k+1-i} = 1/r_i$ and $r_i > 1$. □

A paving matroid of rank at least two is loopless, and under the convention of [7] that Y_M equals the inverse Z-polynomial of the matroid obtained by deleting the loops, every matroid of rank zero has $Y_M = 1$ and every matroid of rank one has $Y_M = 1 + t$; hence Conjectures 1.4 and 1.5 of [7] hold for all paving matroids.

Strict log-concavity cannot be strengthened to real-rootedness: already $Y_{U_{3,4}}(t) = (1+t)(3+5t+3t^2)$ has two nonreal roots.

Outline of the proof. Let $k \geq 2$. By the formula of Gao, Li and Xie, Y_M is $Y_{U_{k,n}}$ minus one correction polynomial for each hyperplane with at least

k elements. In a paving matroid every $(k-1)$-subset of the ground set lies in exactly one hyperplane. Dividing by $\binom{n}{k-1}$ therefore writes the coefficient sequence of Y_M as a convex combination of an explicit sequence v, coming from $U_{k,n}$, and explicit sequences w_h, one for each hyperplane size h; the weight of w_h is the proportion of $(k-1)$-subsets covered by hyperplanes of size h (Section 3). Positive mixtures of log-concave sequences need not be log-concave. Here, however, the consecutive ratios of all components in a given position lie in a common interval, and these intervals are pairwise separated (Section 4). This proves all inequalities in odd rank (Section 5). In even rank $k = 2m$ the central coefficient of each correction contains an extra term, and the inequality at index $m-1$ needs a quantitative lower bound on a_{m-1}. We obtain it by noting that at most one hyperplane can be large, and that the weight of a large hyperplane is bounded in terms of its size (Section 6). Remark 7.2 shows that a single component w_h need not be log-concave; the argument compensates for this by using the actual weights of the components.

Dependencies. Besides standard facts on matroids and Möbius functions, the proof uses three results from the literature that we do not reprove: the formula for $Y_{U_{k,n}}$ [7, Theorem 1.1], the palindromicity of Y_M [7, Lemma 2.2], and the formula for paving matroids [6, Theorem 1.7]. The latter is equivalent to [7, Proposition 5.3]; as explained after Theorem 2.6, this follows by applying both formulas to $U_{k-1,h} \oplus U_{1,1}$ and using the multiplicativity of Y [7, Proposition 2.3]. The equivalence is not needed in the proof. The nonnegativity of the coefficients of Q_M [2] is used only in Remark 2.3. Theorem numbers of [7] and [6] refer to the arXiv versions arXiv:2507.01332v1 and arXiv:2601.17314v1, and the page number of [4] to arXiv:2212.03190v3.

Organization. Section 2 collects definitions and the results we use. Section 3 sets up the convex decomposition, Section 4 proves the ratio bounds, and Sections 5 and 6 treat odd and even rank. Section 7 completes the proof and contains remarks on rank two and on the role of the hyperplane weights.

2. Preliminaries

2.1. **Matroids.** We follow Oxley [9]. Let M be a matroid on a finite set E with rank function rk and closure operator cl. Its flats form a geometric lattice $\mathcal{L}(M)$. For a flat F we write $M|F$ for the restriction to F and M/F for the contraction; if M is loopless, so are $M|F$ and M/F, and $\mathcal{L}(M|F)$ and $\mathcal{L}(M/F)$ are the intervals below and above F in $\mathcal{L}(M)$. The Möbius invariant of M is $\mu_M = \mu_{\mathcal{L}(M)}(\mathrm{cl}(\emptyset), E)$. If M is loopless, then $\mu_M \neq 0$ and

$$(-1)^{\mathrm{rk}\, M}\mu_M = |\mu_M| > 0 \tag{2.1}$$

by Rota's sign theorem for geometric lattices [11].

Let M have rank k. A *hyperplane* is a flat of rank $k-1$. Following Ferroni, Nasr and Vecchi [5], a hyperplane H is *stressed* if every k-element subset of H is a circuit. The uniform matroid $U_{k,n}$ has rank k and $n \geq k$ elements, every k-subset being a basis.

Lemma 2.1. *Let M be a paving matroid of rank $k \geq 2$ on a set E of n elements.*

(a) *Every hyperplane with at least k elements is stressed.*
(b) *Every $(k-1)$-subset of E lies in exactly one hyperplane.*
(c) *Distinct hyperplanes H, H' satisfy $|H \cap H'| \leq k-2$ and $|H|+|H'| \leq n+k-2$.*

Consequently, if λ_h denotes the number of hyperplanes of M with h elements, then

$$\sum_{h=k}^{n-1} \lambda_h \binom{h}{k-1} \leq \binom{n}{k-1}. \tag{2.2}$$

Proof. Since every circuit has at least k elements, every subset of E with at most $k-1$ elements is independent. (a) A k-subset X of a hyperplane is dependent, and each of its proper subsets is independent; hence X is a circuit. (b) A $(k-1)$-subset I is independent, so $\operatorname{cl}(I)$ is a hyperplane containing I. A hyperplane $H \supseteq I$ contains $\operatorname{cl}(I)$ and has the same rank, so $H = \operatorname{cl}(I)$. (c) By (b), $H \cap H'$ contains no $(k-1)$-subset, so $|H \cap H'| \leq k-2$, and then $|H| + |H'| = |H \cup H'| + |H \cap H'| \leq n+k-2$. Finally, by (b) the sets of $(k-1)$-subsets contained in the various hyperplanes are pairwise disjoint, which gives (2.2). □

A hyperplane is a proper subset of E, so hyperplanes have at most $n-1$ elements. By Lemma 2.1(a), for a paving matroid λ_h ($h \geq k$) is the number of stressed hyperplanes with h elements, which is the quantity appearing in [6, 7]. A paving matroid of rank at least two is loopless.

2.2. **The inverse Z-polynomial.** Let M be loopless. We write $Q_M(t)$ for the inverse Kazhdan–Lusztig polynomial normalized as in [8, Section 2.2]: the polynomial $(-1)^{\operatorname{rk} M} Q_M(t)$ is the value at $(\operatorname{cl}(\emptyset), E)$ of the inverse, in the incidence algebra of $\mathcal{L}(M)$ over $\mathbb{Z}[t]$, of the Kazhdan–Lusztig function. With this normalization Q_M has nonnegative coefficients [2]. Since every Kazhdan–Lusztig polynomial has constant term 1, setting $t = 0$ turns the Kazhdan–Lusztig function into the zeta function of $\mathcal{L}(M)$, whose inverse is the Möbius function; hence $Q_M(0) = (-1)^{\operatorname{rk} M} \mu_M = |\mu_M|$.

Definition 2.2 ([7])**.** Let M be a loopless matroid of rank k. The *inverse Z-polynomial* of M is

$$Y_M(t) = (-1)^k \sum_{F \in \mathcal{L}(M)} (-1)^{\operatorname{rk} F} Q_{M|F}(t)\, t^{\,k - \operatorname{rk} F}\, \mu_{M/F}. \tag{2.3}$$

This is [7, equations (2) and (3)]. A related polynomial appears earlier in [4, p. 19, footnote 7]; throughout we use the normalization (2.3) of [7]. For matroids with loops one sets $Y_M = Y_{M'}$, where M' is obtained by deleting the loops [7]; we only need loopless matroids. Since $\operatorname{rk}(M/F) = k - \operatorname{rk} F$,

(2.1) turns (2.3) into

$$Y_M(t) = \sum_{F \in \mathcal{L}(M)} |\mu_{M/F}|\, t^{\,k-\operatorname{rk} F}\, Q_{M|F}(t). \tag{2.4}$$

Remark 2.3. For every loopless matroid M of rank k all coefficients of Y_M in degrees $0, \dots, k$ are positive. Indeed, every summand of (2.4) has non-negative coefficients by [2]. Given $0 \le i \le k$, let I be a set of $k-i$ elements of a basis and $F = \operatorname{cl}(I)$, a flat of rank $k-i$. The summand of F contributes $|\mu_{M/F}|\, Q_{M|F}(0) = |\mu_{M/F}|\, |\mu_{M|F}| > 0$ to the coefficient of t^i. This argument is not needed below: for paving matroids positivity also follows from the proof of Theorem 1.1.

Lemma 2.4 ([7, Lemma 2.2]; see also [2, p. 5])**.** *For every matroid M, $t^{\operatorname{rk} M} Y_M(t^{-1}) = Y_M(t)$.*

Theorem 2.5 ([7, Theorem 1.1])**.** *Let $s \ge k \ge 1$, and write $Y_{U_{k,s}}(t) = \sum_{i=0}^{k} u_i(s) t^i$. Then $u_{k-i}(s) = u_i(s)$ for $0 \le i \le k$, and*

$$u_i(s) = \binom{s}{i}\binom{s-i-1}{s-k} \qquad \Big(0 \le i \le \Big\lfloor \frac{k}{2} \Big\rfloor\Big). \tag{2.5}$$

For instance, a loopless matroid of rank one on n elements is $U_{1,n}$, and $Y_{U_{1,n}}(t) = 1 + t$.

Theorem 2.6 ([6, Theorem 1.7])**.** *Let M be a paving matroid of rank k on n elements, and let λ_h be the number of stressed hyperplanes of M with h elements. Then*

$$Y_M(t) = Y_{U_{k,n}}(t) - \sum_{h=k}^{n-1} \lambda_h\, c_{k,h}(t), \tag{2.6}$$

where $c_{k,h} = Y_{U_{k,h}}$ if k is odd, and, if $k = 2m$ is even,

$$c_{k,h}(t) = Y_{U_{k,h}}(t) - d_h\, t^m, \tag{2.7}$$

where

$$d_h = \frac{4h}{(2h-2m)(2h-2m+2)} \binom{h-1}{m,\, m-1,\, h-2m}.$$

Here $\binom{h-1}{m,\, m-1,\, h-2m} = \frac{(h-1)!}{m!\,(m-1)!\,(h-2m)!}$ is a multinomial coefficient. In [6] the correction term is written with the factor $1 + (-1)^k$, which vanishes for odd k. For $k \ge 2$ and $h \ge k$ one also has

$$c_{k,h} = Y_{U_{k,h+1}} - (1+t)\, Y_{U_{k-1,h}}, \tag{2.8}$$

which is the correction term of [7, Proposition 5.3]. Indeed, let H be the ground set of $U_{k-1,h}$. The circuits of $U_{k-1,h} \oplus U_{1,1}$ are the k-subsets of H, so it is a paving matroid of rank k on $h+1$ elements. The set H is a hyperplane with $h \ge k$ elements, and every other hyperplane consists of the coloop and $k-2$ elements of H. So H is its only hyperplane with at least k elements, and (2.6) gives $Y_{U_{k-1,h} \oplus U_{1,1}} = Y_{U_{k,h+1}} - c_{k,h}$. On the other

hand $Y_{U_{k-1,h}\oplus U_{1,1}} = (1+t)Y_{U_{k-1,h}}$ by multiplicativity [7, Proposition 2.3] and $Y_{U_{1,1}} = 1+t$. Since both (2.6) and [7, Proposition 5.3] subtract from $Y_{U_{k,n}}$ one correction for each stressed hyperplane, depending only on k and its size, (2.8) shows that the two formulas agree. We shall use (2.7). A short computation puts d_h in the form

$$d_h = \binom{h}{2m-1}\binom{2m-1}{m}\frac{h-2m+1}{(h-m)(h-m+1)}. \tag{2.9}$$

Indeed,

$$\frac{4h\,(h-1)!}{(2h-2m)(2h-2m+2)} = \frac{h!}{(h-m)(h-m+1)},$$
$$\frac{h!}{m!\,(m-1)!\,(h-2m)!} = \binom{h}{2m-1}\binom{2m-1}{m}(h-2m+1).$$

3. A convex decomposition

Throughout Sections 3–6, M is a paving matroid of rank $k \geq 2$ on n elements, λ_h is the number of hyperplanes with h elements, and $Y_M(t) = \sum_i a_i t^i$. Put

$$B = \binom{n}{k-1}, \qquad b_h = \binom{h}{k-1}, \qquad \alpha_h = \frac{\lambda_h b_h}{B} \quad (k \leq h \leq n-1),$$

and $\alpha_0 = 1 - \sum_{h=k}^{n-1}\alpha_h$. By (2.2) all α's are nonnegative, and they sum to one. We also write

$$\ell = \Big\lfloor \frac{k-1}{2} \Big\rfloor,$$

so that $\ell = m$ if $k = 2m+1$ and $\ell = m-1$ if $k = 2m$; the indices $0 \leq i \leq \ell$ are those strictly below $k/2$.

Lemma 3.1. *For $s \geq k$ and $0 \leq i \leq \lfloor k/2 \rfloor$,*

$$f_s(i) := \frac{u_i(s)}{\binom{s}{k-1}} = \binom{k-1}{i}\frac{s-k+1}{s-i}.$$

Proof. We have

$$\binom{s-i-1}{s-k} = \binom{s-i-1}{k-1-i} = \frac{s-k+1}{s-i}\binom{s-i}{k-1-i},$$
$$\binom{s}{i}\binom{s-i}{k-1-i} = \binom{s}{k-1}\binom{k-1}{i};$$

now use (2.5). □

For $0 \leq i \leq k$ put

$$v_i = \frac{u_i(n)}{B}, \qquad z_i = \frac{a_i}{B},$$

and for $k \leq h \leq n-1$ and $0 \leq i \leq k$ put

$$w_{h,i} = v_i - \frac{[t^i]\,c_{k,h}}{b_h}.$$

Thus $v_i = f_n(i)$ for $i \le \lfloor k/2 \rfloor$. Since $Y_{U_{k,n}}$ and $Y_{U_{k,h}}$ are palindromic of degree k and the extra term in (2.7) sits in the middle degree, we have $v_{k-i} = v_i$ for $0 \le i \le k$, and $w_{h,k-i} = w_{h,i}$ for each h as above. Theorem 2.6 and $\alpha_0 + \sum_h \alpha_h = 1$ give

$$z_i = \alpha_0 v_i + \sum_{h=k}^{n-1} \alpha_h \, w_{h,i} \qquad (0 \le i \le k). \tag{3.1}$$

If $n = k$, the sum is empty and $z_i = v_i$. We refer to v and the w_h as the *components*; the component w_h only matters when $\alpha_h > 0$.

Lemma 3.2. (a) *For $0 \le i \le \ell$ and every h with $k \le h \le n-1$,*

$$w_{h,i} = (k-1)\binom{k-2}{i}\frac{n-h}{(n-i)(h-i)} > 0. \tag{3.2}$$

(b) *If $k = 2m$, then $v_m - v_{m-1} = \delta$, where*

$$\delta = \binom{2m-1}{m}\frac{n-2m+1}{(n-m)(n-m+1)} > 0, \tag{3.3}$$

and $w_{h,m} - w_{h,m-1} = \delta$ for every h with $k \le h \le n-1$.

Proof. (a) For $i \le \ell$ we have $[t^i]c_{k,h} = u_i(h)$: for odd k by the definition $c_{k,h} = Y_{U_{k,h}}$, and for $k = 2m$ by (2.7), since $i < m$. Hence $w_{h,i} = f_n(i) - f_h(i)$. By Lemma 3.1,

$$\begin{aligned} w_{h,i} &= \binom{k-1}{i}\frac{(n-k+1)(h-i) - (h-k+1)(n-i)}{(n-i)(h-i)} \\ &= \binom{k-1}{i}\frac{(n-h)(k-1-i)}{(n-i)(h-i)}, \end{aligned}$$

and $\binom{k-1}{i}(k-1-i) = (k-1)\binom{k-2}{i}$. All factors are positive because $i \le k-2 < h < n$.

(b) Put $C = \binom{2m-1}{m} = \binom{2m-1}{m-1}$. For $s \ge 2m$, Lemma 3.1 gives

$$\begin{aligned} f_s(m) - f_s(m-1) &= C(s-2m+1)\Big(\frac{1}{s-m} - \frac{1}{s-m+1}\Big) \\ &= \frac{C(s-2m+1)}{(s-m)(s-m+1)}. \end{aligned}$$

For $s = n$ this is δ. For $s = h$ it equals d_h/b_h by (2.9). Hence

$$\frac{[t^m]\, c_{k,h}}{b_h} = f_h(m) - \frac{d_h}{b_h} = f_h(m-1) = \frac{[t^{m-1}]\, c_{k,h}}{b_h},$$

and therefore $w_{h,m} - w_{h,m-1} = v_m - v_{m-1} = \delta$. Finally $n \ge 2m$ gives $\delta > 0$. □

Corollary 3.3. *$z_i > 0$ for $0 \le i \le \lfloor k/2 \rfloor$. If $k = 2m$, then $z_m = z_{m-1} + \delta$.*

Proof. For $i \le \ell$ every component is positive by Lemmas 3.1 and 3.2(a), and the weights are nonnegative with sum one. If k is odd, then $\lfloor k/2 \rfloor = \ell$. If $k = 2m$, then $z_m = z_{m-1} + \delta$ by (3.1) and Lemma 3.2(b), again because the weights sum to one. □

4. Separated ratio intervals

For $1 \le i \le \ell$ define

$$L_i = \frac{(k-1-i)(n-i+1)}{i\,(n-i-1)}, \qquad U_i = \frac{(k-i)(n-i+1)}{i\,(n-i)}. \tag{4.1}$$

These are well defined and positive: $i \le \ell \le k-2$, and $n-i-1 \ge k-\ell-1 \ge 1$.

Lemma 4.1. *Let $1 \le i \le \ell$. Then*

$$L_i \le \frac{v_i}{v_{i-1}} = U_i,$$

and for every h with $k \le h \le n-1$,

$$L_i \le \frac{w_{h,i}}{w_{h,i-1}} < U_i.$$

Moreover $L_i > U_{i+1}$ for $1 \le i \le \ell - 1$.

Proof. By Lemma 3.1 and (3.2),

$$\frac{v_i}{v_{i-1}} = \frac{k-i}{i} \cdot \frac{n-i+1}{n-i} = U_i, \qquad \frac{w_{h,i}}{w_{h,i-1}} = \frac{k-1-i}{i} \cdot \frac{n-i+1}{n-i} \cdot \frac{h-i+1}{h-i}.$$

A direct computation gives

$$U_i - L_i = \frac{(n-i+1)(n-k)}{i\,(n-i)(n-i-1)} \ge 0.$$

Next,

$$\frac{w_{h,i}/w_{h,i-1}}{U_i} = \frac{(k-1-i)(h-i+1)}{(k-i)(h-i)} < 1$$

because $(k-i)(h-i) - (k-1-i)(h-i+1) = h-k+1 > 0$, and

$$\frac{w_{h,i}/w_{h,i-1}}{L_i} = \frac{(h-i+1)(n-i-1)}{(h-i)(n-i)} \ge 1$$

because $(h-i+1)(n-i-1) - (h-i)(n-i) = n-h-1 \ge 0$. Finally, for $i \le \ell - 1$,

$$\frac{L_i}{U_{i+1}} = \frac{(i+1)(n-i+1)}{i\,(n-i)} > 1.$$ □

The following observation is the reason the components can be combined.

Lemma 4.2. *Let $x^{(1)}, \dots, x^{(r)}$ be sequences, let $\beta_1, \dots, \beta_r \ge 0$ be weights, not all zero, and let $y = \sum_j \beta_j x^{(j)}$. If, for some index i and some $0 < L \le U$, every $x^{(j)}$ with $\beta_j > 0$ satisfies $x^{(j)}_{i-1} > 0$ and $L \le x^{(j)}_i / x^{(j)}_{i-1} \le U$, then $y_{i-1} > 0$ and $L \le y_i / y_{i-1} \le U$.*

Proof. The quotient y_i/y_{i-1} is the average of the ratios $x_i^{(j)}/x_{i-1}^{(j)}$ with the nonnegative weights $\beta_j x_{i-1}^{(j)}$, not all zero. □

Corollary 4.3. *For $1 \le i \le \ell$ we have $L_i \le z_i/z_{i-1} \le U_i$, and*

$$z_i^2 > z_{i-1}z_{i+1} \qquad (1 \le i \le \ell - 1).$$

Proof. The first assertion follows from (3.1) and Lemmas 4.1 and 4.2. For $1 \le i \le \ell - 1$ it gives $z_{i+1}/z_i \le U_{i+1} < L_i \le z_i/z_{i-1}$, and all terms are positive. □

5. Odd rank

Proposition 5.1. *If $k = 2m+1 \ge 3$, then $a_i^2 > a_{i-1}a_{i+1}$ for $1 \le i \le k-1$.*

Proof. Here $\ell = m$. Since $a_i = a_{k-i}$ by Lemma 2.4, the inequality at an index i is equivalent to the inequality at $k-i$; as $i \le m$ or $k-i \le m$ for every $1 \le i \le k-1$, it suffices to treat $1 \le i \le m$. For $1 \le i \le m-1$ this is Corollary 4.3. By Corollary 4.3 with $i = m$,

$$\frac{z_m}{z_{m-1}} \ge L_m = \frac{m\,(n-m+1)}{m\,(n-m-1)} = \frac{n-m+1}{n-m-1} > 1.$$

Hence $a_m > a_{m-1}$, and since $a_{m+1} = a_m$,

$$a_m^2 - a_{m-1}a_{m+1} = a_m(a_m - a_{m-1}) > 0.$$

In rank three ($m = 1$) the range $1 \le i \le m-1$ is empty; the index $i = 1$ is the case $i = m$ just treated, and $i = 2$ follows by symmetry. □

6. Even rank

Let $k = 2m \ge 2$, so that $\ell = m - 1$. Put

$$N = n - m, \qquad D = N - m + 1 = n - 2m + 1, \qquad C = \binom{2m-1}{m},$$

so that $N \ge m$, $D \ge 1$ and, by (3.3), $\delta = CD/(N(N+1))$. By Corollary 3.3, $z_m = z_{m-1} + \delta > z_{m-1}$; together with $a_{m+1} = a_{m-1}$ this gives the inequality at the middle index,

$$a_m^2 - a_{m-1}a_{m+1} = a_m^2 - a_{m-1}^2 > 0. \tag{6.1}$$

For $m = 1$ this is the only internal index. From now on let $m \ge 2$, and put

$$p = z_{m-1}, \qquad q = z_{m-2}.$$

By symmetry, the inequality at index $m+1$ is the same as the one at index $m-1$, and the latter reads

$$p^2 > q\,(p + \delta). \tag{6.2}$$

6.1. **The two coefficients below the middle.** For real u with $m-1 \le u \le N-1$ put

$$\pi(u) = \frac{m\,(N-u)}{(N+1)(u+1)}, \qquad \kappa(u) = \frac{(m-1)(N-u)}{(N+2)(u+2)}.$$

The function π is positive and strictly decreasing. Attach to v the parameter $u = m-1$ and to w_h the parameter $u = h-m$; the latter ranges over $m \le u \le N-1$.

Lemma 6.1. *For every component $x \in \{v\} \cup \{w_h : 2m \le h \le n-1\}$ with parameter u,*

$$x_{m-1} = C\,\pi(u), \qquad x_{m-2} = C\,\kappa(u).$$

Consequently $q < \eta\, p$, where

$$\eta = \frac{(m-1)(N+1)}{m\,(N+2)} < 1.$$

Proof. For w_h use (3.2) together with $(2m-1)\binom{2m-2}{m-1} = mC$ and $(2m-1)\binom{2m-2}{m-2} = (m-1)C$, and note $n-h = N-u$, $n-m+1 = N+1$, $h-m+1 = u+1$, $n-m+2 = N+2$, $h-m+2 = u+2$. For v use Lemma 3.1 with $\binom{2m-1}{m-1} = C$ and $\binom{2m-1}{m-2} = \frac{m-1}{m+1}C$:

$$v_{m-1} = \frac{CD}{N+1} = C\pi(m-1), \qquad v_{m-2} = \frac{(m-1)CD}{(m+1)(N+2)} = C\kappa(m-1).$$

For every u we have $\kappa(u)/\pi(u) = \eta\,(u+1)/(u+2) < \eta$, so every component satisfies $x_{m-2} < \eta\, x_{m-1}$; averaging with the weights of (3.1) gives $q < \eta p$. □

6.2. **A lower bound for the coefficient below the middle.**

Lemma 6.2. *For real numbers $a \ge s \ge 1$ and $b \ge 1$,*

$$\frac{s}{a+2b+(2b+1)s}\Big(\frac{a+2b}{a+b} + \frac{(2b+1)a}{s+b}\Big) > \frac{a+s-1}{a+s+2b+1}. \tag{6.3}$$

Proof. All denominators are positive. Multiplying by them, (6.3) becomes $\Phi > 0$, where

$$\begin{aligned}\Phi = {} & s\big[(a+2b)(s+b) + (2b+1)a(a+b)\big](a+s+2b+1)\\ & - (a+s-1)(a+b)(s+b)\big[a+2b+(2b+1)s\big].\end{aligned}$$

Substituting $a = s+x$ and expanding gives $\Phi = \Phi_0 + \Phi_1 x + \Phi_2 x^2 + \Phi_3 x^3$ with

$$\begin{aligned}\Phi_0 &= 2(s+b)\big((b+1)s+b\big)(2s+b),\\ \Phi_1 &= 4b(s^3-1) + (6b^2+2b+6)(s^2-1)\\ &\qquad + 2b(b^2-b+4)(s-1) + 7b^2+14b+6,\\ \Phi_2 &= 6b(s^2-1) + (4b^2-2b+2)(s-1) + b^2+5b+2,\\ \Phi_3 &= b(2s-1).\end{aligned}$$

For $s \geq 1$ and $b \geq 1$ every summand displayed in $\Phi_0, \ldots, \Phi_3$ is nonnegative and the last one in each line is positive; since $x = a - s \geq 0$, we get $\Phi > 0$. $\square$

Proposition 6.3. *Let $k = 2m \geq 4$. Then*

$$p \geq \frac{C\, m\, D}{(N+1)(N+m+1)}. \tag{6.4}$$

Proof. By (3.1) and Lemma 6.1, p/C is a weighted average of the values $\pi(u)$ over the parameters u of the components, with the weights α_0, α_h. Call a hyperplane H *large* if $2|H| > n + 2m - 2$. By Lemma 2.1(c) there is at most one large hyperplane, and if H is large, every other hyperplane H' has $|H'| \leq n + 2m - 2 - |H|$.

Case 1: no hyperplane is large. Then every hyperplane with at least k elements has parameter $u = h - m \leq (n-2)/2 = (N+m-2)/2$, and so does v, since $m - 1 \leq (N+m-2)/2$ is equivalent to $m \leq N$. As π is decreasing,

$$\frac{p}{C} \geq \pi\Big(\frac{N+m-2}{2}\Big) = \frac{m\,(D+1)}{(N+1)(N+m)} > \frac{m\,D}{(N+1)(N+m+1)},$$

where the last inequality is $(D+1)(N+m+1) - D(N+m) = N+m+1+D > 0$. This includes the case $\lambda_h = 0$ for all h.

Case 2: there is a large hyperplane H, with $h = |H|$ elements. Any other hyperplane with h elements would also be large, so $\lambda_h = 1$, and the weight of w_h is $\alpha := \alpha_h = b_h/B$. Put

$$s = n - h \geq 1, \qquad a = h - 2m + 2, \qquad b = m - 1 \geq 1.$$

Largeness of H means $a > s$, and $h = a+2b$, $n = a+s+2b$, $N = a+s+b-1$, $D = a + s - 1$. Every other hyperplane H' with at least k elements has parameter $|H'| - m \leq s + m - 2$, and so does v, because $s \geq 1$. Since π is decreasing,

$$\frac{p}{C} \geq \alpha\, \pi(h-m) + (1-\alpha)\, \pi(s+m-2), \tag{6.5}$$

where

$$\pi(h-m) = \frac{m}{N+1} \cdot \frac{s}{a+b}, \qquad \pi(s+m-2) = \frac{m}{N+1} \cdot \frac{a}{s+b}.$$

We bound $\alpha = \binom{h}{2b+1}/\binom{h+s}{2b+1}$ from above. Each of the $2b+1$ factors in its product form is at most the first factor $h/(h+s)$, and Bernoulli's inequality bounds the resulting power:

$$\alpha = \prod_{j=0}^{2b} \frac{h-j}{h+s-j} \leq \Big(1 + \frac{s}{h}\Big)^{-(2b+1)} \leq \frac{1}{1 + (2b+1)s/h} =: \bar{\alpha}.$$

Since $a > s$, we have $s(s+b) < a(a+b)$, that is, $\pi(h-m) < \pi(s+m-2)$; so the right-hand side of (6.5) decreases when α is replaced by $\bar{\alpha} \geq \alpha$. Using

$h = a + 2b$ this gives

$$\begin{aligned}\frac{p}{C} &\geq \frac{m}{N+1} \cdot \frac{s}{a+2b+(2b+1)s}\Big(\frac{a+2b}{a+b} + \frac{(2b+1)a}{s+b}\Big) \\ &> \frac{m}{N+1} \cdot \frac{a+s-1}{a+s+2b+1}\end{aligned}$$

by Lemma 6.2. Since $a + s - 1 = D$ and $a + s + 2b + 1 = N + m + 1$, this is (6.4). □

Proposition 6.4. *If $k = 2m \geq 2$, then $a_i^2 > a_{i-1}a_{i+1}$ for $1 \leq i \leq k-1$.*

Proof. The middle index is (6.1); this settles $m = 1$. Let $m \geq 2$. By symmetry it suffices to treat $1 \leq i \leq m-1$. For $1 \leq i \leq m-2$ this is Corollary 4.3. For $i = m-1$ we prove (6.2). We have

$$1 - \eta = \frac{N+m+1}{m\,(N+2)}, \qquad \eta\,\delta = \frac{(m-1)\,C\,D}{m\,N\,(N+2)}.$$

By Proposition 6.3,

$$\begin{aligned}(1-\eta)\,p - \eta\,\delta &\geq \frac{CD}{(N+1)(N+2)} - \frac{(m-1)CD}{mN(N+2)} \\ &= \frac{C\,D\,(N-m+1)}{mN(N+1)(N+2)} = \frac{C\,D^2}{mN(N+1)(N+2)} > 0.\end{aligned}$$

Since $q < \eta p$ (Lemma 6.1) and $p + \delta > 0$,

$$p^2 - q\,(p+\delta) > p^2 - \eta p\,(p+\delta) = p\big[(1-\eta)p - \eta\delta\big] > 0. \qquad \square$$

7. Proof of the main theorem and remarks

Proof of Theorem 1.1. If $k = 1$, then $M = U_{1,n}$ and $Y_M(t) = 1 + t$ by Theorem 2.5. Let $k \geq 2$. Symmetry is Lemma 2.4. By Corollary 3.3, $a_i = Bz_i > 0$ for $0 \leq i \leq \lfloor k/2 \rfloor$, and symmetry gives positivity for the remaining indices. Part (ii) is Proposition 5.1 for odd k and Proposition 6.4 for even k. □

Remark 7.1 (Rank two). Every loopless matroid of rank two is paving. If it has $c \geq 2$ parallel classes, (2.4) gives directly

$$Y_M(t) = (c-1) + c\,t + (c-1)\,t^2,$$

since $|\mu_M| = c-1$, $Q_M(t) = c-1$, and each rank-one flat contributes t. The inequality of Theorem 1.1(ii) reads $c^2 - (c-1)^2 = 2c - 1 > 0$.

Remark 7.2 (The hyperplane weights matter). A single component w_h need not be log-concave. The proof compensates for this by using the actual weights of the components, through Proposition 6.3. Let $k = 4$, $n = 8$ and $h = 7$. Then

$$\begin{aligned}(v_0, \ldots, v_4) &= \tfrac{1}{56}(35, 120, 140, 120, 35), \\ c_{4,7}(t) &= 20 + 70t + 70t^2 + 70t^3 + 20t^4,\end{aligned}$$

since $Y_{U_{4,7}} = 20 + 70t + 84t^2 + 70t^3 + 20t^4$ and $d_7 = 14$, and with $b_7 = 35$

$$(w_{7,0}, \dots, w_{7,4}) = \Big(\frac{3}{56}, \frac{1}{7}, \frac{1}{2}, \frac{1}{7}, \frac{3}{56}\Big), \qquad w_{7,1}^2 - w_{7,0}w_{7,2} = -\frac{5}{784}.$$

In an actual paving matroid of rank four on eight elements, a hyperplane with seven elements has weight $\binom{7}{3}/\binom{8}{3} = 5/8$ in (3.1), not 1, and by Lemma 2.1(c) it is the only hyperplane with at least four elements. The resulting matroid is $U_{3,7} \oplus U_{1,1}$, and

$$Y_{U_{3,7}\oplus U_{1,1}}(t) = 15 + 50t + 70t^2 + 50t^3 + 15t^4,$$

in accordance with Theorem 1.1 and with the multiplicativity of Y under direct sums [7, Proposition 2.3].

Remark 7.3. The argument uses the paving hypothesis twice: through the explicit formula of Theorem 2.6, and through Lemma 2.1, which bounds the total weight of the corrections and allows at most one large hyperplane. Beyond paving matroids, the formula for elementary split matroids [7, Proposition 5.2] involves stressed subsets of every rank together with the inverse Z-polynomials of cuspidal matroids, and we do not know whether a similar separation of ratio intervals holds in that setting.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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Email address: `math@knowledgesense.jp`